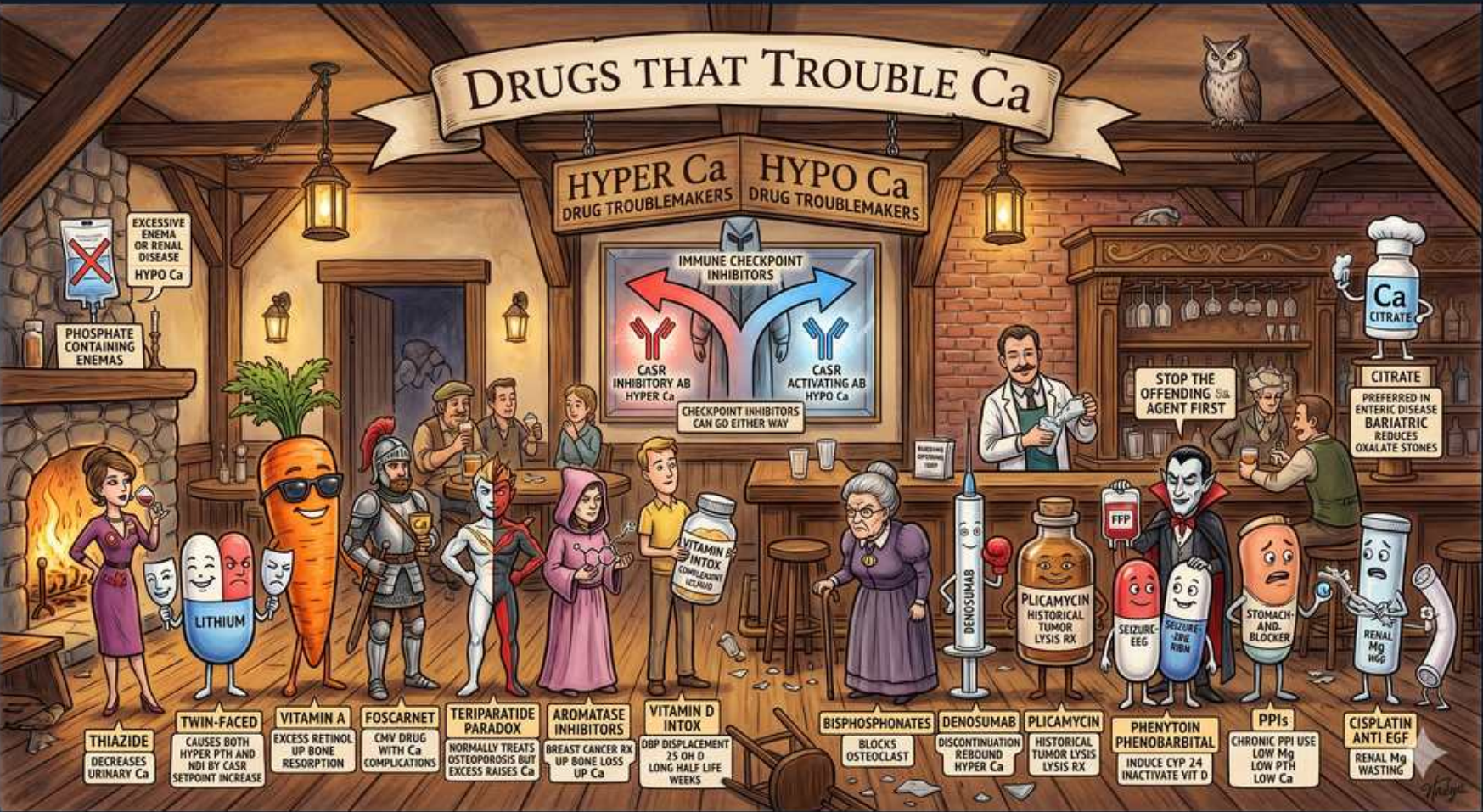


Drugs That Alter Calcium — Hyper- vs Hypocalcemia



1. Thiazides → hypercalcemia

WHAT YOU SEE

Two-faced figure 'THIAZIDE — decrease urinary Ca'

WHAT IT ENCODES

Thiazides increase distal tubular calcium REABSORPTION, lowering urinary Ca and mildly raising serum Ca. Used therapeutically in hypercalciuric stones, but can unmask hyperparathyroidism.

BOARD TRAP

LOOP diuretics do the opposite (calciuric) and are used to treat hypercalcemia — don't confuse loops with thiazides.

2. Lithium → hypercalcemia

WHAT YOU SEE

Pill couple 'LITHIUM'

WHAT IT ENCODES

Lithium shifts the CaSR set-point (mimics FHH), raising PTH and serum calcium. Can cause lithium-induced hyperparathyroidism over time.

BOARD TRAP

Lithium hypercalcemia resembles primary HPT/FHH with high-normal PTH and low urine Ca — review the med list.

3. Vitamin A → hypercalcemia

WHAT YOU SEE

Carrot figure 'VITAMIN A — excess bone resorption'

WHAT IT ENCODES

Vitamin A (and retinoids) in excess stimulate osteoclastic bone resorption → hypercalcemia.

BOARD TRAP

Both vitamin A and vitamin D excess cause hypercalcemia, but D acts via gut absorption and A via bone resorption.

4. Foscarnet → Ca/Mg chelation

WHAT YOU SEE

Knight 'FOSCARNET — DRY GULCH complications'

WHAT IT ENCODES

Foscarnet chelates divalent cations, causing acute symptomatic HYPOcalcemia and hypomagnesemia (and nephrotoxicity).

BOARD TRAP

Foscarnet LOWERS calcium — it's an antiviral cause of hypocalcemia, not hyper.

5. Teriparatide → hypercalcemia

WHAT YOU SEE

Pink-cloaked figure 'TERIPARATIDE — INTERMITTENT PTH ANABOLIC'

WHAT IT ENCODES

Teriparatide (recombinant PTH 1-34) given intermittently is ANABOLIC (builds bone) but can cause transient hypercalcemia. Continuous PTH excess is catabolic.

BOARD TRAP

Intermittent PTH builds bone (osteoporosis Rx); continuous high PTH (hyperparathyroidism) resorbs bone — the dosing pattern flips the effect.

6. Aromatase inhibitors → bone loss

WHAT YOU SEE

Veiled figure 'AROMATASE INHIBITORS — direct cancer Rx, low bone'

WHAT IT ENCODES

Aromatase inhibitors (anastrozole, letrozole) deplete estrogen → accelerated bone loss/osteoporosis. Used in ER+ breast cancer.

BOARD TRAP

Als harm bone (need DEXA/bisphosphonate); tamoxifen is comparatively bone-protective in postmenopausal women.

7. Vitamin D intoxic → hypercalcemia

WHAT YOU SEE

Bottle 'VITAMIN D INTOX — 25 D long half life'

WHAT IT ENCODES

Vitamin D toxicity raises gut calcium absorption → hypercalcemia, hypercalciuria, nephrocalcinosis. 25-OH-D has a long half-life so it persists for weeks.

BOARD TRAP

Treat with hydration and glucocorticoids; effects are prolonged because of 25-D's long half-life.

8. Bisphosphonates → hypocalcemia

WHAT YOU SEE

Syringe 'BISPHOSPHONATES — BLOCKS OSTEOCLAST'

WHAT IT ENCODES

Bisphosphonates inhibit osteoclast-mediated resorption (treat hypercalcemia of malignancy, osteoporosis, Paget). Can cause hypocalcemia, jaw osteonecrosis, atypical femur fractures.

BOARD TRAP

Correct vitamin D/calcium deficiency before bisphosphonates or denosumab to avoid severe hypocalcemia.

9. Denosumab → hypocalcemia

WHAT YOU SEE

'DENOSUMAB — RANKL antibody, hypocalcemia risk'

WHAT IT ENCODES

Denosumab is a RANKL monoclonal antibody blocking osteoclast formation. Risk of hypocalcemia (esp. in CKD), jaw osteonecrosis; rebound vertebral fractures on stopping.

BOARD TRAP

Unlike bisphosphonates, denosumab is not renally cleared (usable in CKD) but raises hypocalcemia risk in low GFR — check Ca/vitamin D first.

10. Plicamycin → hypocalcemia

WHAT YOU SEE

Vampire figure 'PLICAMYCIN — historical, fixes hyperCa'

WHAT IT ENCODES

Plicamycin (mithramycin) is a historical osteoclast inhibitor once used for hypercalcemia of malignancy; largely replaced by bisphosphonates/denosumab.

BOARD TRAP

Mostly of historical/board interest — modern first-line for malignant hypercalcemia is IV fluids + calcitonin + bisphosphonate.

11. Phenytoin/phenobarbital → low D

WHAT YOU SEE

Bottles 'PHENYTOIN PHENOBARBITAL — induce CYP, low vit D, low Ca'

WHAT IT ENCODES

Enzyme-inducing antiepileptics accelerate vitamin D catabolism via CYP induction → osteomalacia and hypocalcemia with chronic use.

BOARD TRAP

Chronic anticonvulsants cause hypocalcemia/osteomalacia through vitamin D loss — supplement vitamin D.

12. PPIs → low Ca absorption

WHAT YOU SEE

Pill 'PPIs — chronic, low Ca absorption'

WHAT IT ENCODES

Chronic PPI use reduces calcium absorption (needs gastric acid) and is associated with increased fracture risk and hypomagnesemia.

BOARD TRAP

PPI-related hypomagnesemia can itself cause refractory hypocalcemia — replace magnesium first.

13. Citrate → hypocalcemia

WHAT YOU SEE

'Ca CITRATE — preferred in renal disease/bariatric, reduces stones'

WHAT IT ENCODES

Citrate chelates ionized calcium — massive blood transfusion or regional citrate dialysis can cause acute hypocalcemia. Calcium citrate salt is preferred for absorption in achlorhydria/bariatric patients.

BOARD TRAP

Citrate toxicity (transfusion) lowers ionized Ca causing perioral tingling — give IV calcium.

14. Cisplatin/anti-EGF → hypomagnesemia

WHAT YOU SEE

Figure 'CISPLATIN ANTI-EGF — low Mg'

WHAT IT ENCODES

Cisplatin and anti-EGFR agents (cetuximab) waste renal magnesium → hypomagnesemia, which impairs PTH secretion/action and causes secondary hypocalcemia.

BOARD TRAP

Hypocalcemia refractory to calcium = check magnesium; replace Mg to restore PTH responsiveness.

15. Checkpoint inhibitors — either way

WHAT YOU SEE

Center panel 'IMMUNE CHECKPOINT INHIBITORS — CaSR can go either way'

WHAT IT ENCODES

Immune checkpoint inhibitors can cause hypophysitis/hypoparathyroid or, conversely, granulomatous reactions, so they can produce either hypo- or hypercalcemia.

BOARD TRAP

Don't assume a single direction with checkpoint inhibitors — endocrinopathies are variable; check PTH and Ca.

16. Phosphate enemas → hypocalcemia

WHAT YOU SEE

'EXCESSIVE ENEMA — PHOSPHATE-CONTAINING ENEMAS, HYPO Ca'

WHAT IT ENCODES

Phosphate-containing enemas/bowel preps can cause acute hyperphosphatemia that precipitates calcium → acute hypocalcemia (and nephrocalcinosis), especially in CKD/elderly.

BOARD TRAP

Hyperphosphatemia from phosphate preps drives Ca down acutely — avoid in renal impairment.